

Pushpak Vijay Katkhede

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SUMMARY

Software Engineer with 1+ year of experience building LLM-powered AI systems and 3+ years of software engineering experience across backend, distributed, and cloud-native systems. Delivered RAG pipelines, agent orchestration workflows, and LLM-assisted systems at Oregon State University, alongside production backend services and distributed systems at IBM. Published author at ICSE 2026 (ACM/IEEE) on NLP-based REST API defect analysis. Open to remote and/or relocation.

TECHNICAL SKILLS

Languages: Python, Bash, Go

AI/ML: LLMs, RAG, NLP, Embeddings, Prompt Engineering, LLM Evaluation, LLM-as-a-Judge, Evaluation Harnesses, TensorFlow, Human-in-the-Loop, PyTorch, scikit-learn, XGBoost, Transformers, Fine Tuning

LLM & AI Frameworks: LangChain, LangGraph, LangSmith, Ollama, Anthropic Claude, OpenAI, MCP, Tool Calling, Hugging Face, Vector Search, Hybrid Search, Reranking, Guardrails

Backend: FastAPI, Node.js, Express.js, REST APIs, Kafka, Microservices, Distributed Systems, JSON, XML, Unit Testing, Redis

Cloud & DevOps: Docker, Kubernetes, Jenkins, Terraform, CI/CD, Git, Linux, AWS, Microsoft Azure, GitHub Actions

Databases: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, AWS DynamoDB), Vector Databases (ChromaDB, Pinecone)

WORK EXPERIENCE

Oregon State University

Remote

Software Engineer

May 2025 - Present

- Developed a stateful LLM-assisted account compromise detection system processing 500K+ daily Duo MFA and Microsoft Entra ID logs, combining deterministic rule-based filtering, agent-orchestrated investigation, structured LLM triage, and human-in-the-loop escalation.
- Built an LLM-powered identity resolution system across dual systems of record, resolving 4,500+ duplicate identities and 100+ new identities daily using phonetic blocking, fuzzy candidate generation, LLM-based identity equivalence scoring, and confidence-driven human review.
- Engineered 20+ Python and SQL integration pipelines synchronizing data attributes across enterprise systems and mapping 850+ access configuration groups across 8+ applications, de-risking a large-scale HR platform migration from on-prem to cloud.
- Delivered an internal tool and its reusable Python SDK for multi-platform API consumption, authentication, logging, automating daily operations and cutting development effort by 50%.

Research Assistant

Dec 2024 - May 2025

- Formulated API defect categorization as an NLP classification problem, defining a ground-truth taxonomy of bug types across 200K+ GitHub issues to serve as labeled training data for downstream BERT embedding and LLM-based models.
- Designed and implemented a scalable research pipeline for bug extraction using GitHub API mining over 200K issues, integrating custom tokenization, BERT embeddings, and StarCoder2 LLMs for representation learning and topic modeling.

IBM India

Bangalore, KA, IN

Software Engineer

Oct 2019 - Jul 2022

- Designed and built a self-service trial environment for observability pipelines with automated infrastructure provisioning using Terraform, enabling 50+ enterprise customers to run evaluations that converted into \$3M in signed contracts.
- Led a cross-functional team of 3 interns to design and build a suite of simulators for API interaction, microservice failures, and user behavior simulation enabling fault-injection for assessment of the complex observability pipeline.
- Modernized a large-scale legacy web application from vanilla JavaScript to React, implementing component-based architecture serving live charts and dashboards improving data refresh latency to under 2 seconds.
- Improved application performance by 30% by refactoring inventory, purchase order, and workflow management services in a MERN application using event-driven architecture, asynchronous programming, and caching.
- Containerized and orchestrated 15+ microservices serving 100k+ users, integrating them into CI/CD pipelines with automated deployment and achieving 99% fault-tolerant deployments with monitoring dashboards.

PROJECTS

ARGUS - Autonomous Reasoning Gateway for Unified Systems (Python, Claude API, Kubernetes, LangSmith)

- Engineered an end-to-end LLM-powered AIOps agent using the Anthropic Claude API with structured tool calling, orchestrating diagnostic workflows across Prometheus, Splunk, and kubectl to detect, diagnose, and remediate incidents, with a human-in-the-loop approval guardrail before executing remediation actions.
- Deployed a real-time observability and telemetry pipeline feeding metrics, logs, and alert signals into the LLM inference loop; tracked diagnostic runs using LangSmith, logging tool-call traces, LLM-as-a-Judge scores, and root-cause accuracy against incident ground truth to benchmark prompt and model changes across incidents.

Ask chAMP! - RAG Chatbot (Python, OpenAI API, ChromaDB, BM25, FastAPI, Kafka)

- Developed a production RAG chatbot with multi-source ingestion and multi-query hybrid retrieval using vector search and BM25, applying Reciprocal Rank Fusion reranking(RRF) and exposing FastAPI REST APIs for real-time query and response generation under 10 seconds.
- Built a Kafka-based event-driven incremental ingestion pipeline processing 4.5GB+ across 3k+ PDFs, GitHub repositories, and SharePoint documents using context-aware semantic chunking and incremental embeddings, enabling near-real-time ChromaDB updates and GPT-4o mini responses (87% LLM-judged groundedness) with persistent context and caching.

PUBLICATIONS

- Defects4REST: A Benchmark of Real-World Defects to Enable Controlled Testing and Debugging Studies for REST APIs**, ACM/IEEE International Conference on Software Engineering (ICSE), 2026.

EDUCATION

Master of Science, Computer Science (AI Focus)

Sep 2022 - Dec 2024

Oregon State University

Corvallis, OR, US

- GPA:** 3.75/4.0

- Coursework:** Algorithms Design and Analysis, Software Engineering, Parallel Programming, Deep Learning, Project Management

Bachelor of Engineering, Computer Engineering

Aug 2015 - Jun 2019

Savitribai Phule Pune University

Pune, MH, IN

- GPA:** 8.10/10.0

- Coursework:** Data Structures, Database Management Systems, Operating Systems, Machine Learning, Software Design and Modeling